BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI (RAJ)
I SEMESTER 2025-2026
ASSIGNMENT

Course No.:	**CS/SS G527**	**Course Title:**	**Cloud Computing**
Deadline:	**As on Nalanda**	**Maximum Marks:**	**60 (15%)**

Note:

- **Group of maximum four students** **Group sheet with domains**
- **Submit deliverables on Nalanda (only one per group)**
- **Demonstration will be conducted after submissions.**
- **All members of a group should divide the tasks (a-h) equally i.e. 2 tasks per each member and implement 2 services by each member. Each student should demonstrate their tasks and answer questions. 12M is reserved for individual contribution.**

In this problem, you will implement a cloud application with at least 8 microservices (as given against your chosen scenario in the sheet) using any public cloud provider environment, out of which at least one being a web service accessible over public URL. The application you build for the scenario is mentioned against groups at this link. Chose your group to know the domain to work with.

Following requirements must be met.

a. All cloud infrastructure (networks, clusters, databases, storage, IAM policies, etc.) across all cloud providers must be provisioned exclusively using an **Infrastructure as Code** (IaC) platform (e.g., Terraform).

b. Make one of the 8 services as **serverless function** (e.g., AWS Lambda, Google Cloud Functions, or Azure Functions) to perform an asynchronous, event-driven task. Microservices must communicate using cloud-native mechanisms: REST, gRPC, or message queues (e.g., Kafka, AWS SQS, Google Pub/Sub). There should be use of each of these communication mechanisms.

c. Use a managed **K8s service** (e.g., EKS / GKE / AKS) to host stateless microservices. Implement **Horizontal Pod Autoscalers** (HPAs) for at least two critical services, configured to scale based on CPU or memory utilization. Use a load testing tool (e.g., k6, JMeter, Gatling) to validate your system's resilience and scalability. **Load Test**: Generate sustained traffic to demonstrate the HPA scaling out your services.

d. All Kubernetes deployments and application updates must be managed via a **GitOps controller** (e.g., ArgoCD, Flux). The controller must track a Git repository. Direct kubectl apply is forbidden for service deployment.

e. One of 8 services must be a real-time stream processing service (**Flink**) running on the **managed cluster** (e.g., Google Dataproc, AWS EMR, or Azure Synapse/HDInsight). This service must consume event from a **Kafka** topic. It must perform a stateful, time-windowed aggregation (e.g., "count of unique users per 1-minute window"). The aggregated results must be published back to a separate 'results' Kafka topic. The Kafka cluster itself may be a managed cloud service (e.g., MSK, Confluent Cloud, Aiven).

f. For the concept declared along with your scenario, think how this concept can be applied to your domain, and implement it. Demo should include it.

g. Use at least three distinct cloud storage products. For example: an **object store** (e.g., S3, GCS) for raw data (e.g., file uploads that trigger the serverless function), a **managed SQL database** (e.g., RDS, Cloud SQL) for relational data (e.g., user accounts, structured metadata), a **managed NoSQL database** (e.g., DynamoDB, Firestore, MongoDB Atlas) for high-throughput, semi-structured data (e.g., session state, real-time analytics results).

h. Implement a comprehensive observability stack. **Metrics**: Deploy Prometheus and Grafana. Create a dashboard showing key service metrics (RPS, error rate, latency) and Kubernetes cluster health. **Logging**: Implement a centralized logging solution (e.g., EFK stack, Loki, or Cloud Logging) to aggregate logs from all microservices, including the analytics job.

Deliverables:

- Design document containing System overview, Cloud deployment architecture, Microservices architecture digrams, microservice responsibilities, interconnection mechanisms, and the rationale behind design choices in the cloud.
- Microservices code (in Github repository), all IaC script code, all Kubernetes manifests and GitOps configuration
- task_divison.txt file should contain idno, name of all members and tasks assigned to each member.

[60M]

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